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Palaeomagnetism of Late Triassic volcanic rocks from the western margin of Khorat Basin, Thailand and its implication for ambiguous inclination shallowing in Mesozoic sediments of Indochina

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SUMMARY

Palaeomagnetic constraints are essential factors in the reconstruction of the Mesozoic convergence of Eastern Asia blocks. As one of the key blocks, Indochina was constrained only by sedimentary-rocks-derived palaeomagnetic data. To evaluate whether the palaeomagnetic data used to restore the Late Triassic position of Indochina suffered inclination shallowing effects, we conducted a palaeomagnetic and geochronologic study on a coeval volcanic clastic rocks sequence in the western margin of the Khorat Basin, Thailand. The U-Pb SIMS dating on zircons indicates the age of the sampling section is between 205.1 ± 1.5 and 204.7 ± 1.4 Ma. Site mean directions are $D_g/I_g = 217.2°/-39.4°$ ($\kappa_g = 45.1$, $\alpha_{95g} = 10.1°$) before and $D_s/I_s = 209.2°/-44.5°$ ($\kappa_s = 43.8$, $\alpha_{95s} = 10.2°$) after tilt correction. The new data set indicates a positive reversal test result at 'Category C' level. The characteristic remanent magnetization recorded by the coexistent magnetite and hematite is interpreted to be primary remanence acquired during the initial cooling of the volcanic clastic rocks. The consistence of the corresponding palaeolatitudes derived from the volcanic clastic rocks and the former reported sedimentary rocks suggests that there is probably no significant inclination shallowing bias in the sedimentary-rocks-derived palaeomagnetic data. Therefore, the estimates of the Late Triassic position of Indochina are confirmed to be reliable. The Indochina Block had collided to the southern margin of Eurasia by the Late Triassic and played an important role in the Mesozoic convergence of the Eastern Asia blocks.

Key words: Asia; Palaeomagnetism; Rock and mineral magnetism; Continental tectonics: strike-slip and transform.

1 INTRODUCTION

Palaeomagnetic data can provide a quantitative estimate on the rotation around vertical axes and latitudinal tectonic displacement during tectonic reconstructions. As one of the key blocks making up the Eastern Asia, Indochina (or Annamia) is considered to drift from the South Hemisphere to North Hemisphere during the late Palaeozoic to early Mesozoic, and its Mesozoic amalgamating history has been constrained palaeomagnetically (Barr & Macdonald 1979; Achache & Courtillot 1985; Maranate & Vella 1986; Yang & Besse 1993; Richter & Fuller 1996; Singsoupho *et al.* 2014; Yan *et al.* 2017). In the tectonic application of palaeomagnetism, the magnetic inclination is very useful to calculate the palaeolatitude for the unit of measuring ($\tan I = 2\tan \lambda$, where I is the magnetic inclination and λ is palaeolatitude), based on the geocentric axial dipole hypothesis model.

However, a variety of studies on sedimentary rocks have documented shallowing bias of the magnetic inclinations (e.g. Kent & Smethurst 1998; Gilder *et al.* 2003; Tan *et al.* 2007). Different shallowing factors f ($\tan I_o = f * \tan I_f$, where f is the shallowing factor, I_o is the observed inclination, I_f is the inclination of geomagnetic field direction; King 1955) were recognized from different sedimentary units, which can be lower than 0.5 (Gilder *et al.* 2001; Dupont-Nivet *et al.* 2002; Ding *et al.* 2015) or larger than 0.8 (Wang &

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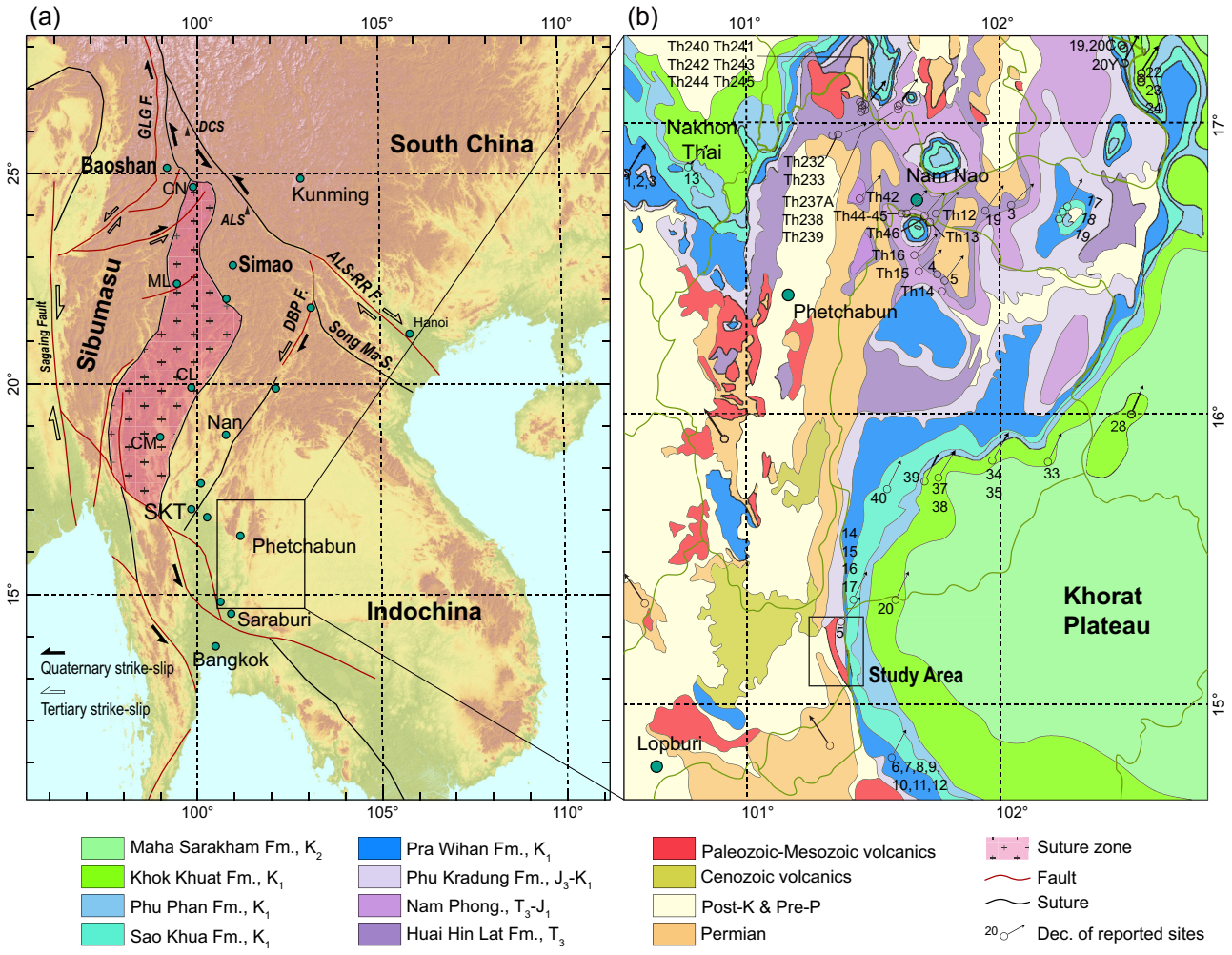


Figure 1. (a) Schematic tectonic map of Southeast Asia; CN (Changning), ML (Menglian), CL (Chianglai), CM (Chiangmai), SKT (Sukhothai), ALS-RR F. (Ailao Shan-Red River Fault), DBP F. (Dien Bien Phu Fault), GLG F. (Gaoligong Fault), DCS (Dianchang Shan), ALS (Ailao Shan); (b) Geological map of study area (revised from 1:5000 000 geological map of Thailand, DMR, 2011; Ridd *et al.* 2011) and reported palaeomagnetic sites in this area, details are summarized in Supporting Information Table S2.

Yang 2007; Zhao *et al.* 2017), while some other authors obtained sediment-derived results that may not suffer the problem (Li *et al.* 2013). The inclination shallowing bias is hotly debated for decades because it may lead unacceptable errors in the palaeolatitude estimation. But till now, we reached no consensus on questions such as are all sedimentary rocks suffered inclination shallowing effect? What are the dominant factors for inclination shallowing, compaction degree, magnetic mineral state, geographic position during forming or some other unknown causes? To restore the real magnetic data, two independent statistical or experimental approaches have been developed to detect and correct the inclination shallowing bias, and their assumptions as well as advantages and shortcomings are quite different from each other. The magnetic anisotropy method (Jackson *et al.* 1991; Kodama 1997; Tan *et al.* 2002) analyses individual magnetic particle anisotropy on a relatively small subset and extrapolate to the total specimens involved. The elongation/inclination method assumes that the origins of the data set scatter are limited to be geomagnetic (palaeosecular variation) and sedimentary flattening (Tauxe & Kent 2004), and will be non-effective for small data set ($N < 100$). Correction methods can provide (semi-)quantitative estimates on inclination shallowing problem, but the use of the corrections on sedimentary rocks is frequently unavailable because of the exacting or impractical process. Obtaining data from coeval volcanic rocks is always an independent and better approach to evaluate the inclination shallow problem of sediments-derived palaeomagnetic data, though the former may suffer geomagnetic palaeosecular variation problem.

For the Indochina Block, previous reported Mesozoic palaeomagnetic data were revealed mainly from sedimentary rocks, especially the Khorat Group in Khorat Basin of eastern Thailand (Maranate & Vella 1986; Yang & Besse 1993; Bhongsuwan & Elming 2000; Charusiri *et al.* 2006; Takemoto *et al.* 2009; Singsoupho *et al.* 2014; Yan *et al.* 2017), however, no detection on inclination shallowing problem was carried out. To check if the sedimentary rocks in the Khorat Basin suffered significant inclination shallowing and if the shallowing factor ($f = 0.6$) suggested by Torsvik *et al.* (2012) is feasible for the Mesozoic sedimentary rocks of the Indochina Block, we conducted a palaeomagnetic and geochronologic study on a volcanic clastic rock unit in the western margin of Khorat Basin of Thailand. Comparison of data from volcanic and sedimentary rocks

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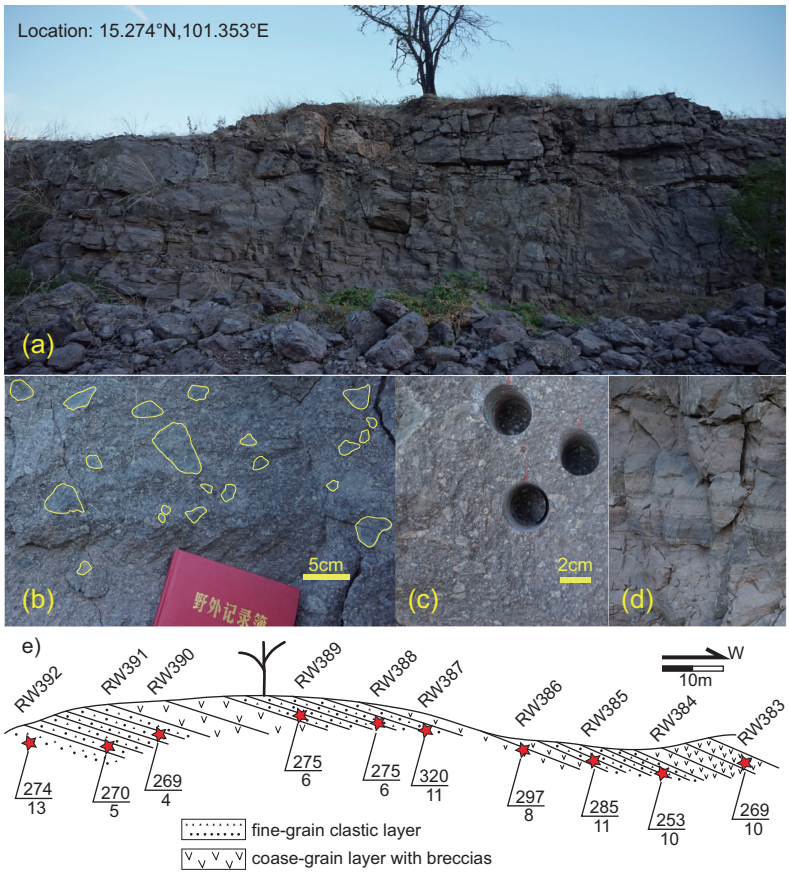


Figure 2. (a) Sampling section; (b) layers with coarse grains; (c,d) layers with fine grains; (e) sampling details as well as the bedding attitudes, beddings of fine grains or welded breccias-tuff are shown.

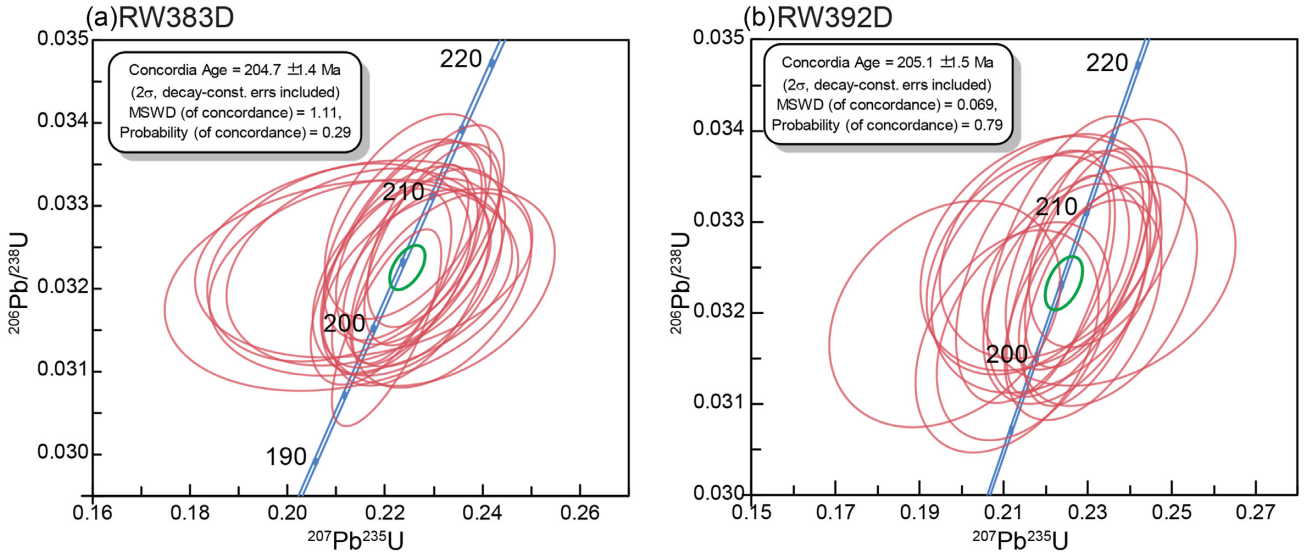


Figure 3. (a and b) Results of SIMS dating on zircons from samples RW383D and RW392D.

is made to evaluate whether the Late Triassic sedimentary rocks of Khorat Group experienced magnetic inclination shallowing.

2 GEOLOGICAL SETTINGS AND SAMPLING

The Southeast Asia Peninsula, comprising Myanmar, Thailand, Laos, Cambodia and Vietnam, is also named as the Indochina Peninsula by Europeans since the Age of Exploration, because it locates between India and China. The Indochina Block, in the sense of geology, occupies the eastern half of the Southeast Asia Peninsula and bounded with the Sibumasu Block to the west, with the South China Block to the northeast by the Changning–Menglian–Chianglai–Chiangmai suture zone and Song Ma suture, respectively

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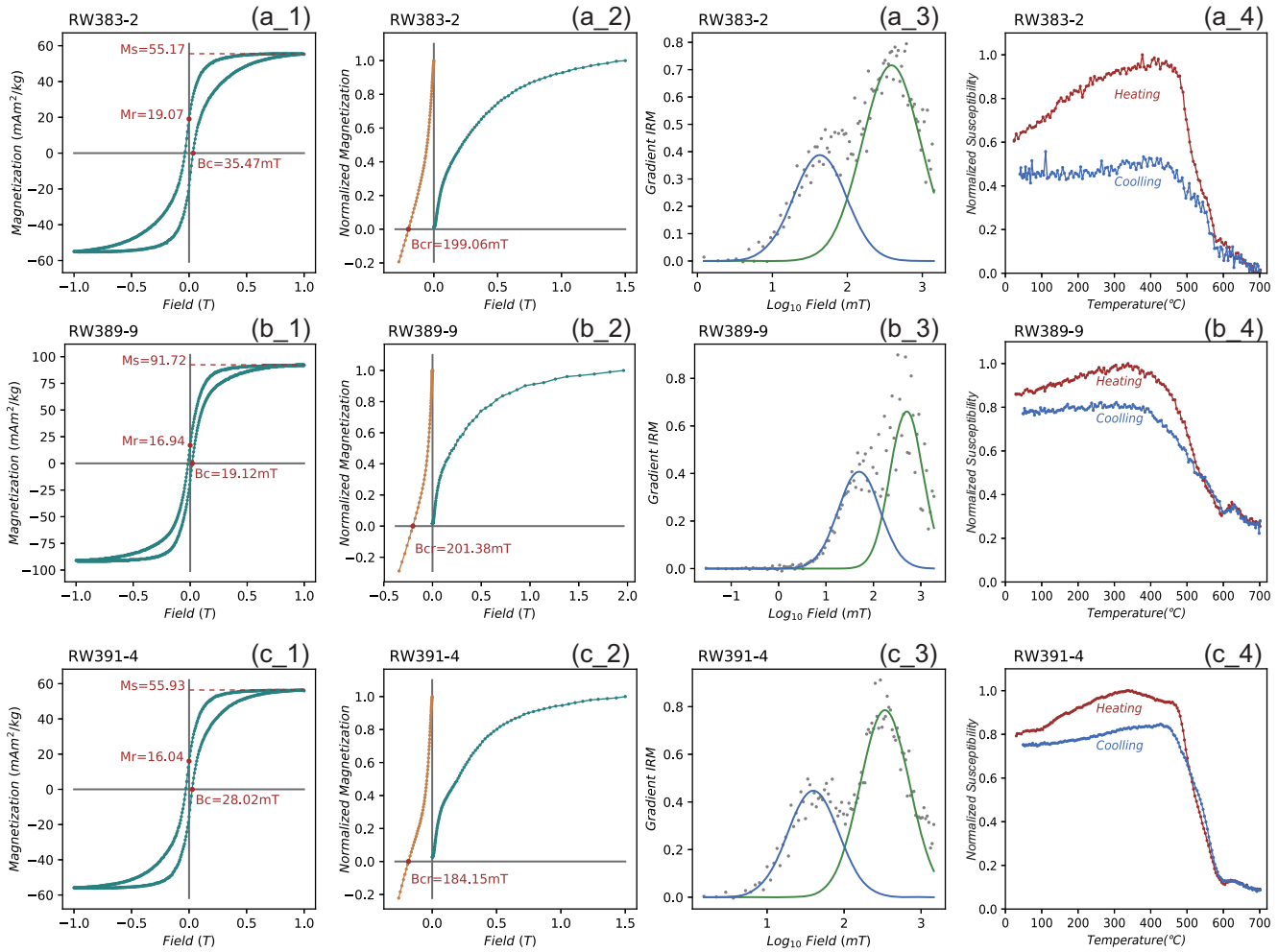


Figure 4. Rock magnetic analyses with hysteresis loops (a_1, b_1, c_1), acquisition and back-field demagnetization (a_2, b_2, c_2) as well as the gradient (a_3, b_3, c_3) of IRMs and thermal dependence of susceptibility curves (a_4, b_4, c_4).

(Ueno 2003; Sone & Metcalfe 2008; Fig. 1a). The central Indochina Block were covered by Mesozoic continental sediments, the Khorat Group. The Khorat Group was subdivided into the Huai Hin Lat, Nam Phong, Phu Kradung, Phra Wihan, Sao Khua, Phu Phan and Khok Kruat formations in ascending stratigraphic order, ranging from the Norian Age of Late Triassic to the Early Cretaceous. The Huai Hin Lat Formation consisted of sandstone, shale and limestone, while the other six formations are mainly sandstones and siltstones. In particular, the Mesozoic strata are well outcropped in the western margin of the Khorat Basin of Thailand and contributed to most of the previously reported palaeomagnetic data from Indochina (Fig. 1b; Yan *et al.* 2017).

Volcanic rocks are widely distributed spatially and temporally in the western margin of Khorat Basin of Thailand and termed as the Loei–Phetchabun Volcanic Belt (Fig. 1b). Petrological, geochemical and geochronological (Rb-Sr, ^{40}Ar-^{39}Ar, U-Pb) studies were reported from some of these rocks and provided preliminary and important information (Barr & Charusiri 2011). The section we chose to collect samples is a volcanic rocks sequence well outcropped during the construction of a railway (Fig. 2). These rocks have a volcanic texture (Figs 2b,c), in which rubbles and breccias can be as large as several centimetres and generally have condensing rims formed during cooling from high temperature. The texture clearly shows that the volcanic rubbles and breccias were welded with matrices (tuff) in a hot condition. The sequence overlies the Middle Permian limestone and is overlain by the Phu Kradung Formation (Late Jurassic to Early Cretaceous), indicated by the contact relations in the field. Its field contact relations to the Huai Hin Lat (Norian Age of Late Triassic) and Nam Phong (Rhaetian Age of Late Triassic to Early Jurassic) formations in the field are not clear because the latter two formations outcropped rarely in this area (Fig. 1b).

At least three stages of volcanic eruptions were included in the sampling section based on the coarse-fine grain circles (Figs 2a,b,c,d,e), with the aim to cover a long enough period so as to average out the geomagnetic secular variation. Core samples were drilled using a gasoline-powered hand-drill, typically 2.5 cm in diameter and 5–8 cm in length. 8–12 piece of core samples are drilled from one site, and a total of 106 piece of core samples were drilled at 10 sites from the section (Fig. 2e). Every individual core sample was oriented with magnetic compass or both magnetic and sun compass when sunshine was available. Local magnetic declination recognized from 72 orientations in the sampling area is −0°48′, which is in agreement with −0°40′ suggested by the World Magnetic Model 2015 (Chulliat *et al.* 2014; see www.magnetic-declination.com). Aiming to constrain the age, two geochronologic samples were collected from the top (RW383D) and the bottom (RW392D) of the sampling section, respectively.

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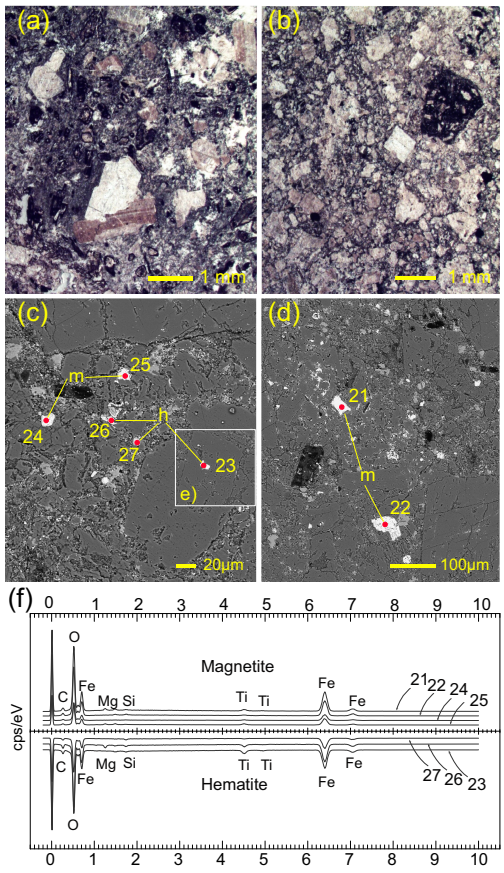


Figure 5. Optical microscope (a and b) and backscatter electron (c and d) images of representative samples; (f) the EDS results of the analysed points in c and d.

3 U-Pb ZIRCON DATING OF VOLCANIC ROCKS

Zircons were extracted from bulk samples of RW383D and RW392D following heavy mineral separation techniques. Zircon grains, together with standard zircon Qinghu, Penglai and Plesovice (Slama *et al.* 2008; Li *et al.*, 2010, 2013), were visually inspected with transmitted and reflected light as well as cathodoluminescence (CL) images after being polished in the epoxy mounts. We used the Cameca IMS-1280 SIMS to analysis U and Pb isotopes in the Institute of Geology and Geophysics, Chinese Academy of Sciences in Beijing. Additional details of analytical procedures are described by Li *et al.* (2009). 20 analyses were subjected to each of the both samples. Data plotting was carried out using the Isoplot-3.6 program (Ludwig 2008). Uncertainties on individual analyses are reported with 2σ in the diagram. As a result, the sampling sequence, from bottom and top, was constrained between 205.1 ± 1.5 Ma and 204.7 ± 1.4 Ma (Fig. 3; Supporting Information Table S1).

4 ROCK MAGNETIC AND PETROGRAPHIC ANALYSES

Powdered representative samples were subjected to rock magnetic analyses with the aim to identify the predominant magnetic carriers as well as their occurrence states. Acquisition and backfield demagnetization of isothermal remanent magnetization (IRM) and hysteresis loops were obtained using a Micro 3900 vibrating sample magnetometer. A mixture of magnetic components with different coercivities (Tauxe *et al.* 1996) was indicated by the wasp-waist hysteresis loops (Figs 4a_1,b_1,c_1). IRMs increase gradually with the applied field and do not reach saturation at 1.5 or 2.0 T (Figs 4a_2,b_2,c_2). Further subsequent coercivity spectra analyses on the IRM acquisition curves (Heslop *et al.* 2002) indicate two components with medium coercivities at ∼30–40 mT and ∼200–300 mT (Figs 4a_3,b_3,c_3). In addition, the thermal dependence of magnetic susceptibility shows the two magnetic components were unblocked at 540–580°C and 630–700°C, respectively (Figs 4a_4,b_4,c_4). Collectively, we interpret the magnetic carriers in the volcanic rocks are both magnetite and hematite.

The coexistence of magnetite and hematite in the volcanic samples was further confirmed in the petrographic analyses.

Observations on thin sections of representative samples reveal that the collected volcanic rocks have a pyroclastic texture (Figs 5a,b). Crystals and rock fragments as well as volcanic glasses distributed randomly in fine-grain matrixes, and showed no significant deformation. Microcrystals, such as feldspar and quartz, keep good crystal form and seem no alteration or metasomatism occurred. No preferred orientation was found, indicating the rocks were not deformed by strain. To further identify the micro-texture and occurrence of magnetic minerals, detailed microscopic examinations were subjected to carbon-coated thin sections, using back-scattering electron (BSE) imaging model equipped with electron microprobe analyser (EMPA) in the lab of the School of Earth Sciences and Engineering of Sun Yat-sen University. Energy dispersive spectrometer (EDS) measurements were conducted on the EMPA (Oxford instruments PLC) at a 15 kV voltage and 9.1 mm working distance.

Most iron oxide minerals revealed in the BSE images are irregular particles (Figs 5c,d). Large grains can be up to 70 μm in size and fine grains distributed randomly. The light of the EDS analyses (Fig. 5f) shows chemical predominance of Fe and O as well as a little Ti, Mg and Si, the iron oxide minerals were identified to be magnetite (analysis points 24, 25 in Fig. 5c; 21–22 in Fig. 5d) and hematite (analysis points 23 in Fig. 5e; 26, 27 in Fig. 5c). Oxidized rims are not recognized in most large grains of magnetite and hematite, though occurred in some fine grains, suggesting that both the magnetite and hematite were mainly crystallized during the initial cooling. The interpretation is consistent with demagnetization results, in which the predominant magnetite and hematite were found to carry the same remanent magnetization (Fig. 6).

5 PALAEOMAGNETIC ANALYSIS

Core samples were trimmed into standard cylindrical specimens, ∼2.2 cm in length. All specimens were thermal demagnetized stepwisely up to 700 °C and measured using a 2G-755 cryogenic magnetometer installed in a magnetically shielded room (residual field weaker than 300 nT) in the Palaeomagnetism and Geochronology Laboratory of the Institute of Geology and Geophysics, Chinese Academy of Science. Characteristic remanent magnetization (ChRM) component was obtained only from the layers consisted of fine grains (Figs 2c,d,e). The samples of site RW383, RW385, RW386 and RW392 from coarse grain layers (Figs 2b,e) behave unstably during thermal demagnetization and yield no meaningful demagnetized results. ChRMs isolated from fine grain samples are of dual polarity (e.g. normal polarity shown in Fig. 6a; reversal polarity shown in Fig. 6c) and calculated using both principle component analysis (PCA) (Figs 6a,c,e,g; Kirschvink 1980) and great circle (GC; Figs 6b,d,f,h; McFadden & McElhinny 1988) methods.

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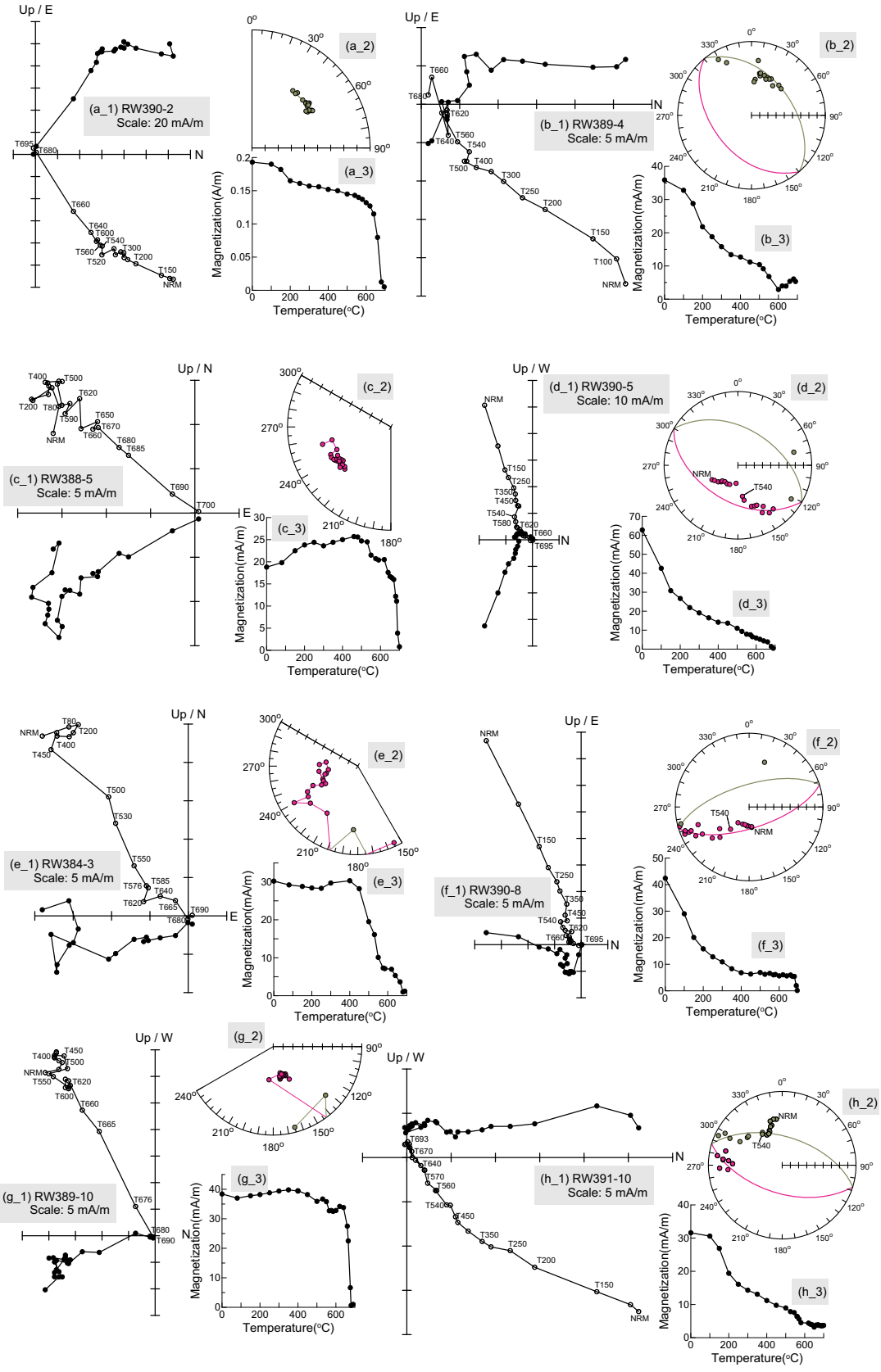


Figure 6. Demagnetization diagrams of representative samples. The solid (open) circles represent horizontal (vertical) projections in Zijderveld diagrams. The green (purple) circles represent vector projection in lower (upper) hemisphere in the equal-area projection diagrams. All diagrams are plotted in geographic coordinate.

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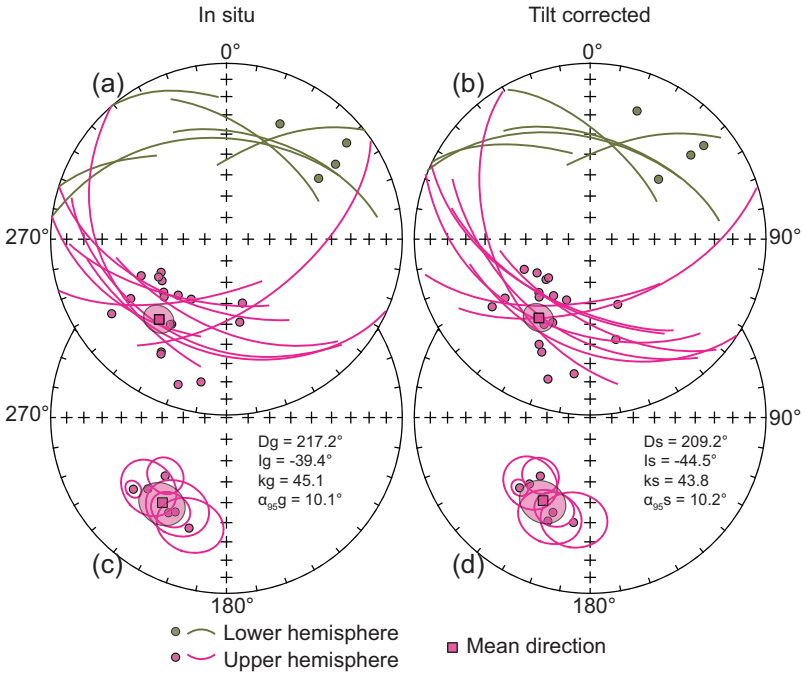


Figure 7. Equal-area projection of the directions of the isolated ChRMs before (a and c) and after (b and d) tilt correction on sample (a and b) and site (c and d) levels.

Table 1. Summary of characteristic remanence from the Late Triassic volcanic rocks.

ID	N	D_g(°)	I_g(°)	D_s(°)	I_s(°)	α_{95}/A_{95} (°)	λ_p(° N)	φ_p(° E)
RW384	5	227.8	−39.4	222.4	−47.6	12.6	48.7	165.0
RW387	5	208.5	−38.8	202.2	−36.6	13.5	68.3	174.5
RW388	7	226.4	−50.2	220.4	−54.0	8.4	49.0	155.8
RW389	5	211.3	−36.7	201.4	−41.6	8.3	68.1	164.7
RW390	8	232.4	−33.6	225.1	−42.8	3.8	46.8	171.1
RW391	6	198.7	−34.0	189.0	−39.4	13.8	78.9	150.3
Site mean	6	217.2	−39.4	–	–	10.1	–	–
	–	–	–	209.2	−44.5	10.2	–	–
VGP	–	–	–	–	–	**11.8**	**60.1**	**164.6**

Abbreviations: N: number of samples or sites used in the mean calculation; $\mathbf{D_g, I_g/D_s, I_s}$: declination and inclination before and after; $\boldsymbol{\alpha_{95}}$ and $\boldsymbol{A_{95}}$: precision parameter and the radius of the cone of 95 per cent confidence for mean direction; $\boldsymbol{\lambda_p/\varphi_p}$: latitude and longitude of virtual geomagnetic pole (VGP).

In most cases, medium (520–580 °C) and high (600–700 °C) temperature components carried by magnetite and hematite, respectively, record the same magnetic direction, following removal of low temperature component before 500 °C (Fig. 6). Especially after 540 °C, the ChRMs got free from the interference of low temperature components (Figs 6a_1,d_1,d_2,f_1,f_2,h_2). However, for different specimens, contribution to the ChRM varies between different magnetic minerals. For specimens such as RW390–2, RW390–5 and RW390–8, the ChRMs are mainly carried by hematites (Figs 6a_3,d_3,f_3). The ChRMs of specimens such as RW388–5 and RW389–10 are contributed more by hematites than magnetites (Figs 6c_3,g_3), and the ChRMs of specimens such as RW384–3 and RW391–10 are more contributed by magnetites than hematites (Figs 6e_3,h_3).

Mean directions are calculated on both specimen (Figs 7a,b) and site levels (Figs 7c,d; Table 1), combining PCA- and GC-analysed ChRMs (McFadden & McElhinny 1988). Specimen mean directions are $D_g/I_g = 219.8°/-40.1°$ ($\kappa_g = 14.8$, $\alpha_{95g} = 6.6°$) before and $D_s/I_s = 213.1°/-45.5°$ ($\kappa_s = 14.9$, $\alpha_{95s} = 6.6°$) after tilt correction, while site mean directions are $D_g/I_g = 217.2°/-39.4°$ ($\kappa_g = 45.1$, $\alpha_{95g} = 10.1°$) before and $D_s/I_s = 209.2°/-44.5°$ ($\kappa_s = 43.8$, $\alpha_{95s} = 10.2°$) after tilt correction. Fisher's precision parameters (κ) are similar before and after tilt correction, both on specimen and site levels, because the bedding dips vary little between sites (Fig. 2). Considering the fold test (McElhinny 1964) is not available, reversal test is carried out on specimens. The angular separation of normal ($D_n/I_n = 44.1°/41.4°$, $\kappa_n = 22.9$, $\alpha_{95n} = 8.1°$) and reversed ($D_r/I_r = 209.2°/-48.2°$, $\kappa_r = 13.0$, $\alpha_{95r} = 10.0°$) ChRM is 12.5°, whereas the critical angle is 13.5° for 95 per cent confidence. Thus, the ChRMs show a positive reversal test result at 'Category C' level (McFadden & McElhinny 1990).

On the other hand, as shown in Fig. 1(b) and summarized in Supporting Information Table S2, Jurassic to Early Cretaceous palaeomagnetic data obtained in this area are predominated by normal polarity records (40 normal sites and 1 reversed site), while the data set in this study is mainly of reversed geomagnetic records (28 of 36 specimens show reversed polarity) (Figs 7a and b). The significant difference from the later Jurassic to Early Cretaceous data set provides an evidence that the studied sequence in this study may not suffer a post-Triassic remagnetization. Moreover, data sets with positive fold test results have been obtained from the underlying Huai Hin Lat Formation and the overlying Sao Khua Formation

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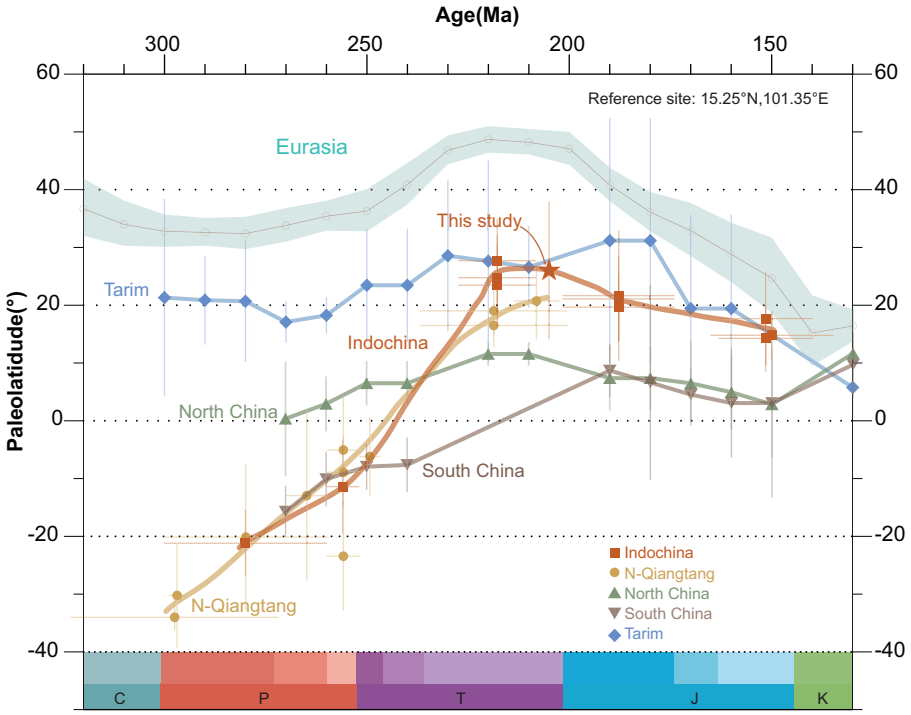


Figure 8. Palaeolatitude evolution of Indochina, Northern Qiangtang, North China, South China, Tarim and Eurasia. Palaeomagnetic data of Indochina and Northern Qiangtang are listed in Table 2, while that of North China, South China, Tarim and Eurasia are listed in Table 3.

(Yang & Besse 1993; Yan *et al.* 2017). These evidences together suggest that the studying area should have escaped from possible Cenozoic remagnetization and then indicates the ChRM preserved in the collected volcanic layers should be acquired before the regional folding.

Collectively, the ChRM carried by the coexistent magnetites and hematites, passed reversal test, is very likely to be a primary geomagnetic record during their initial cooling.

6 DISCUSSIONS

6.1 Palaeoposition of Indochina at the mid-Rhaetian stage

As the latest stage, the Rhaetian stage range from ~208.5 Ma to 201.3 ± 0.2 Ma (Cohen *et al.* 2013). The studying section is between 205.1 ± 1.5 Ma and 204.7 ± 1.4 Ma in age, formed after the Huai Hin Lat Formation and very likely before the Nam Phong Formation. The two formations are Norian of the Late Triassic and Rhaetian of the Late Triassic to Early Jurassic, respectively (Ridd *et al.* 2011). To evaluate whether the sediment-derived palaeomagnetic data from the two formations suffered significant inclination shallowing, their corresponding palaeolatitudes are compared with that of the volcanic-derived value in this study. For the volcanic clastic rocks, site mean VGP (λ_p/φ_p = 60.1°N/164.6°E, A_{95} = 11.8°) indicates a palaeolatitude at 26.4 ± 11.8°N for the reference point 15.25°N/101.35°E. It is similar with the Huai Hin Lat Formation derived palaeolatitudes: 23.9 ± 7.0°N (Achache & Courtillot 1985), 25.2 ± 7.3°N (Yang & Besse 1993) and 28.1 ± 7.2°N (Yan *et al.* 2017), while a little higher than that of Nam Phong Formation derived 22.1 ± 9°N (Maranate & Vella 1986), 21.5 ± 7.3°N (Yang & Besse 1993) and 20.1 ± 5.9°N (Yan *et al.* 2017). As shown in Fig. 8 and Table 2, the variation of palaeolatitudes is compatible with their age, especially the high consistence of the data obtained from the Huai Hin Lat Formation and the new volcanic clastic rocks. Therefore, no significant inclination shallowing bias is recognized from the palaeomagnetic data that was revealed from the Huai Hin Lat Formation limestones and sandstones. Moreover, the consistence of palaeomagnetic data from sedimentary and volcanic rocks proves that the estimations on the Late Triassic palaeoposition of Indochina are reliable.

6.2 Palaeoposition of the Indochina Block in the Mesozoic eastern Asia

To understand the convergence configuration of the Mesozoic Eastern Asia, newly summarized APWP (apparent polar wander path) of key blocks, for example, North China, South China, Tarim, as well as Eurasia are used in the following analyses (Table 3; Fig. 8; Torsvik *et al.* 2012; Huang *et al.* 2018).

Previous reported palaeomagnetic studies have documented that the Indochina Block drifted from the South Hemisphere to North Hemisphere during the late Palaeozoic to early Mesozoic (Table 2; Fig. 8; Yan *et al.* 2018). Recent reported Mesozoic palaeomagnetic data from Indochina indicated a continuous southward motion from *ca.* 33°N to 25°N (for a reference site at 22°N, 102°E) and clockwise rotation (declination from *ca.* 45° to *ca.* 25°) during the Late Triassic to Late Cretaceous (Yan *et al.* 2017). Indochina and Northern Qiangtang collided with Tarim in the Late Triassic, while North China and South China were located at relatively lower palaeolatitudes. Since reliable Late Triassic palaeomagnetic data from South China are still not obtained (Yang & Besse 2001; Huang *et al.* 2018), though previous studies have tried to solve the problem (Zhu *et al.* 1988; Wang 2000), we can just interpret that the collision of North China and South China happened between the Middle Triassic and Early Jurassic based on the decreasing latitudinal difference between the North China and South China blocks as illustrated in Fig. 8. On

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Table 2. Summarized key palaeomagnetic data of the Indochina and Northern Qiangtang blocks.

Rock units	Age	Litho.	λ_s (°N)	φ_s (°E)	N/n	λ_p (°N)	φ_p (°E)	A95 (dp dm^{-1})	F/R test	λ (°)	Reference
Indochina Block											
Phra Wihan-Phu Kradung Fm.	J_3-K_1	Sandstone	16.86	100.48	11/65	60.7	187.4	4	F	15.2	Bhongsuwan & Elming (2000)
Phu Kradung Fm.	J_3-K_1	Sandstone	16.86	100.48	4/29	56.6	181.5	8.0	F	18.1	Bhongsuwan & Elming (2000)
Phu Kradung Fm.	J_3-K_1	Sandstone	16.86	100.48	15/45	58	188	4/8	–	14.7	Maranate & Vella (1986)
Nam Phong Fm.	T_3-J_1	Sandstone	16.67	101.83	8/54	54.4	175.6	7.3	F, R	21.5	Yang & Besse (1993)
Nam Phong Fm.	T_3-J_1	Sandstone	16.67	101.83	9/27	58	174	9/14	–	22.1	Maranate & Vella (1986)
Nam Phong Fm.	T_3-J_1	Sandstone	17.07	101.60	11/75	55.2	178.0	5.9	F, R	20.1	Yan *et al.* (2017)
Volcanics	**~205 Ma**	**Volcanic clastic**	**15.27**	**101.35**	**6/36**	**60.1**	**164.6**	**11.8**	**R**	**26.4**	**This study**
Huai Hin Lat Fm.	T_3	Sandstone, limestone	16.67	101.83	5/56	52.1	169.8	7.3	R	25.2	Yang & Besse (1993)
Huai Hin Lat Fm.	T_3	Sandstone, limestone	16.67–17.05	101.83–101.34	13/124	48.7	165.9	7.2	F, R	28.1	Yan *et al.* (2017)
Huai Hin Lat Fm.	T_3	Sandstone, limestone	16.67	101.83	5/26	49.2	172.3	7.0	F	23.9	Achache & Courtillot (1985)
Can Thuy Fm.	P_3	Basalt	21.38	103.73	4/36	45.6	226.8	8.9	–	−11.7	Chi *et al.* (2016)
Saraburi Group	$P_{1–2}$	Limestone	15.82–14.83	101.35–100.93	15/126	34.1	331.7	5.7	F, R	−21.2	Yan *et al.* (2018)
Northern Qiangtang Block											
Jiapila Fm.	T3	Volcanic rocks	34.10	92.37	29	64	174.7	6.6	F, R	21.0	Song *et al.* (2015)
Tumengela Fm.	T3	Sandstone	33.7	87.8	10	59.2	179.2	4.1	R	19.3	Song *et al.* (2012)
Zhala Fm.	T3	Sandstone	33.1	87.6	11	65.1	184.4	3.7	R	16.7	Song *et al.* (2012)
Kanglu Fm.	T1	Limestone, sandstone	33.7	86.7	2	16.9	202.5	6.7	–	−5.9	Cheng *et al.* (2012)
Rejuechaka Fm.	P3	Limestone, sandstone	33.7	86.7	5	34.4	234.1	6.9/12.5	F, R	−23.1	Cheng *et al.* (2012)
Tuoba Fm.	P3	Siltstone, sandstone	29.7	98.7	10	54.4	231.7	6.1	F	−8.6	Huang *et al.* (1992)
Nayixiong Fm.	P3	Volcanics	34	92	5	11.9	199.7	9.3	F	−4.8	Cheng *et al.* (2013)
Jiushidaoban Fm.	P2	Limestone	34	92	5	−1	204.2	14.6	F	−12.7	Cheng *et al.* (2013)
Changshehu & Xueyuanhe Fm.	P1–2	Limestones	33.7	86.7	8	31.7	226.8	9.2/16.9	F, R	−19.7	Cheng *et al.* (2012)
Nuoribagaribao Fm.	P1	Volcanics	34.1	92.4	14	21.7	232.9	8.9	F, R	−29.8	Song *et al.* (2017)
Zharigen & Nuoribagaribao Fm.	C3–P1	Limestone, sandstone	34.06	92.35	16	25.7	241.5	2.2	C	−33.6	Yang *et al.* (2017)

Abbreviations: N/n: number of sites/samples; $\mathbf{D_s}$, $\mathbf{I_s}$: declination and inclination after tilt correction; $\boldsymbol{\kappa}$ and $\boldsymbol{\alpha_{95}}$: precision parameter and the radius of the cone of 95 per cent confidence for formation mean direction; $\boldsymbol{\lambda_s/\varphi_s}$: latitude and longitude of sampling location; $\boldsymbol{\lambda_p/\varphi_p}$: latitude and longitude of virtual geomagnetic pole (VGP); $\boldsymbol{A_{95}}$**:** the radius of the cone of 95 per cent confidence for formation-mean VGP; **dp/dm**: axes of ellipse of uncertainty of the pole; **F/R test**: Fold or Reversal test; **λ:** Palaeolatitude calculated for the reference point 15.25°N/101.35°E.

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Table 3. Mean poles of the North China, South China, Tarim and Eurasia (Siberia after 250 Ma) as summarized in Huang *et al.* (2018) and Torsvik *et al.* (2012).

AGE (Ma)	No. poles	λ_p (°N)	φ_p (°E)	A_{95}	λ (°)	AGE (Ma)	No. poles	λ_p (°N)	φ_p (°E)	A_{95}	λ (°)	AGE (Ma)	No. poles	λ_p (°N)	φ_p (°E)	A_{95}	λ (°)	AGE (Ma)	λ_p (°N)	φ_p (°E)	A_{95}	λ (°)
North China (Huang *et al.* 2018)						**South China** (Huang *et al.* 2018)						**Tarim** (Huang *et al.* 2018)						**Eurasia** (Torsvik *et al.* 2012)				
130	5	81.5	214.1	7.6	11.8	**130**	13	75.2	210.9	5.5	9.9	**130**	8	67.3	212.8	5.1	6.1	**130**	77.1	182.4	1.3	16.9
150	3	68.8	224.5	15.9	3.1	**150**	5	66.3	219.5	9.3	3.3	**160**	4	60.9	177.9	16.2	19.8	**140**	75.3	181.7	1.3	17.2
160	7	72.8	225.8	6.3	5.2	**160**	5	66.3	219.5	9.3	3.3	**170**	4	60.9	177.9	16.2	19.8	**150**	75.7	162.3	3.7	21.8
170	4	75.8	227.2	7.2	6.7	**170**	1	79.1	226.9	4.5	8.8	**180**	2	52.1	159.1	21.1	31.6	**160**	72.9	138.0	0.6	28.6
180	2	82.0	294.7	5.3	7.5	**180**	2	79.7	246.3	4.5	6.8	**190**	2	52.1	159.1	21.1	31.6	**170**	67.7	117.8	2.2	36.5
190	2	82.0	294.7	5.3	7.5	**190**	1	79.0	225.8	4.5	8.9	**210**	1	52.1	166.8	7	27.0	**180**	68.5	107.2	0.7	36.6
210	2	62.0	7.1	1.9	11.5	**240**	17	44.5	218.2	4.6	−7.3	**220**	2	55.6	163.7	13	28.0	**190**	66.6	101.8	3.3	38.6
220	2	62.0	7.1	1.9	11.5	**250**	25	45.8	219.8	3.9	−7.6	**230**	1	59.0	160.0	13	28.9	**200**	57.7	99.7	1.7	47.5
240	11	58.7	358.4	3.7	6.5	**260**	22	48.6	226.4	4.6	−9.7	**240**	2	52.7	171.8	8	23.9	**210**	53.9	107.4	1.9	51.1
250	11	58.7	358.4	3.7	6.5	**270**	6	53.8	248.1	4.3	−15.3	**250**	2	52.7	171.8	8	23.9	**220**	51.8	120.2	2.6	50.5
260	9	54.2	354.5	4.6	2.9	–	–	–	–	–	–	**260**	8	55.6	180.5	3.1	18.6	**230**	51.3	129.7	0.5	47.4
270	3	50.0	352.9	9.7	0.3	–	–	–	–	–	–	**270**	10	58.5	182.7	3.5	17.5	**240**	54.5	141.8	2.6	39.8
–	–	–	–	–	–	–	–	–	–	–	–	**280**	6	56.0	176.1	10.4	21.1	**250**	56.3	151.4	1.1	34.2
–	–	–	–	–	–	–	–	–	–	–	–	**290**	8	57.6	175.6	7.6	21.2	**260**	54.1	152.2	1.4	34.8
–	–	–	–	–	–	–	–	–	–	–	–	**300**	2	62.4	173.6	15	21.7	**270**	51.5	158.5	0.8	32.1
–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	**280**	46.0	165.4	1.1	28.8
–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	**290**	42.3	167.3	1	27.9
–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	**300**	44.2	166.6	1.6	28.2
–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	**310**	39.7	163.1	4.8	31.3
–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–	**320**	29.6	157.9	1.6	36.3

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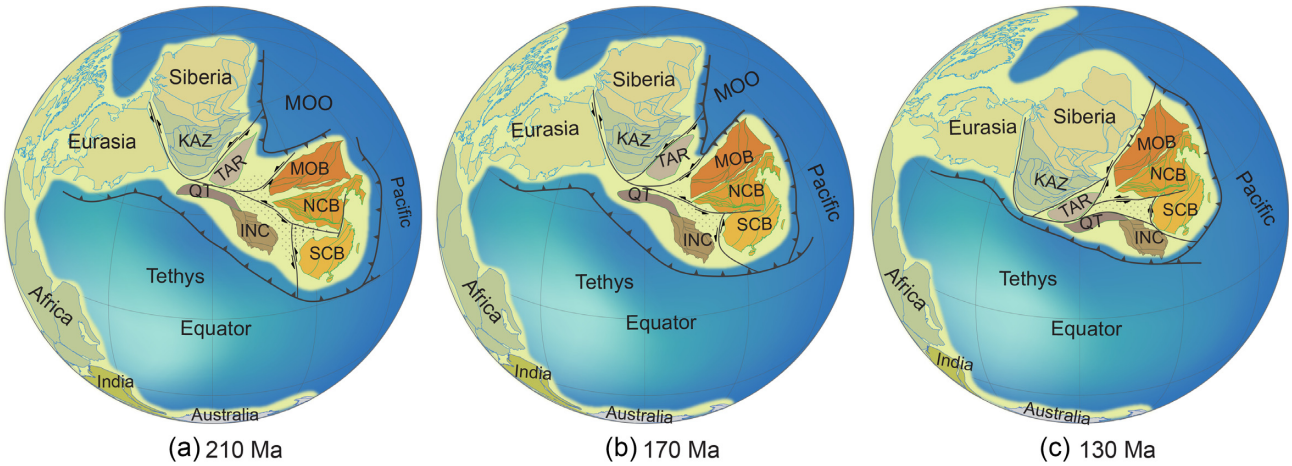


Figure 9. Reconstruction of the evolution of Mesozoic Eastern Asia. INC (Indochina), QT (Northern Qiangtang), SCB (South China Block), NCB (North China Block), MOB (Mongolia Block), TAR (Tarim), KAZ (Kazakhstan), MOO (Mongol-Okhotsk Ocean). MOB merged with NCB in the late Permian (Zhang *et al.* 2018). Strike-slip system is revised from Metelkin *et al.* (2010).

the other hand, Late Triassic sedimentary records in the Longmen Shan area, the western margin of South China, indicate an important tectonic transformation from an early passive continental margin to a foreland basin. Depositional settings changed from marine carbonates to marine clastics and subsequently from marine clastics to continental clastics (Liu *et al.* 2013 and references therein). Therefore, the convergence of North China and South China was very likely started since the Late Triassic, and they merged, as a whole, to Siberia in the latest Jurassic to earliest Cretaceous (Ren *et al.* 2015; Van der Voo *et al.* 2015).

Following the collision between the Northern Qiangtang-Indochina with Tarim in the Late Triassic, a dramatic latitudinal distance of 9–10° was recognized between western blocks (Tarim, Northern Qiangtang, Indochina) and eastern blocks (North China, South China) during the Late Triassic to latest Jurassic (Fig. 8). Inclination shallowing cannot be a reasonable explanation for the gap because the data sets of Tarim and Indochina, as well as the North China and South China all contain data obtained from sedimentary rocks (Yang & Besse 1993; Yan *et al.* 2017; Huang *et al.* 2018). The latitudinal distance indicates a post-Jurassic northward transition of North China–South China relative to Tarim–Northern Qiangtang–Indochina.

6.3 Amount of the extrusion of the Indochina Block

The lateral extrusion model has been widely used to explain the late Mesozoic–Cenozoic tectonic process of Indochina (Tapponnier *et al.* 1982; Yang & Besse 1993; Replumaz *et al.* 2004; Ali & Aitchison 2008; Royden *et al.* 2008; Otofuji *et al.* 2012). The amount of strike-slip along the Ailao Shan-Red River Fault (ASRRF) has long been a hotly debated subject, which has been interpreted to range from ~200–250 km right-lateral to more than 1500 km left-lateral (Dewey *et al.* 1989; Leloup *et al.* 1995).

To dissect the problem, reported palaeomagnetic data with analysing details are compiled (Yang & Besse 1993; Bhongsuwan & Elming 2000; Charusiri *et al.* 2006; Takemoto *et al.* 2009; Otofuji *et al.* 2012; Singsoupho *et al.* 2014; Yan *et al.* 2017). Relative poleward motion between the Indochina and South China blocks after the Early Cretaceous was interpreted to 11.5 ± 6.7° (Yang & Besse 1993), 11.1 ± 6.2° (Singsoupho *et al.* 2014) and 8.9 ± 5.4° (Yan *et al.* 2017), which corresponds a *ca.* 1200–1600 km displacement. Noting that the large displacement estimate is based on the assumption that all the poleward motion was accommodated by the left-lateral displacement along the northwest trending ASRRF.

However, in another aspect, only *ca.* 600–700 km displacement was identified from evidences of left-lateral offsets along the ASRRF (Leloup *et al.* 1995, 2001; Chung *et al.* 1997; Replumaz & Tapponnier 2003; Faure *et al.* 2014). From the study of the Diancang Shan/Ailao Shan metamorphic belt, Tapponnier *et al.* (1990) suggested a ~500 km sinistral displacement across the ASRRF. From the South China and Indochina along the ASRRF, left-lateral offsets of ophiolites and Triassic arc volcanics range between 650–780 km, left-lateral offset of Cretaceous granites is interpreted to be 700 ± 100 km and that of the West Sichuan Basin against West Khorat Basin is 800 ± 200 km (Leloup *et al.* 1995). Chung *et al.* (1997) suggested a ~600 km left-lateral offset based on the correlations of Permian–Triassic flood basalts and late Palaeogene highly potassic mafic magmas from NW Yunnan and NW Vietnam areas.

The dramatic discrepancy of estimates probably implies that not all the poleward motion was accommodated by the strike-slip along the ASRRF. Taking a northwest trending of the ASRRF, the *ca.* 600–700 km left-lateral offset yields merely *ca.* 4–5° poleward motion. The 'missing' amount of poleward motion must be produced in some other process.

From the perspective of mainland Eurasia (Siberia), the South China and North China located at 9–10° latitudinal south to the Tarim–Northern Qiangtang–Indochina during the Late Triassic to latest Jurassic, and then northward moved and merged into mainland Eurasia during the Cretaceous (Fig. 8). In a larger perspective, we interpret that the northward motion of the South China and North China was associated with the closing of the Mongol–Okhotsk Ocean (Ren *et al.* 2015; Van der Voo *et al.* 2015). The combined reconstruction models are shown in Fig. 9. The 'missing' amount is interpreted to be absorbed in the intraplate sinistral strike-slip deformation system developed in the Eastern Asia (Figs 9b,c; Metelkin *et al.* 2007, 2010, 2012; Ren *et al.* 2015), which would not be recorded in the ASRRF. As illustrated in Fig. 9, large sinistral strike-slip faults, for example, Tancheng Lujiang Fault, Longmen Shan faulting belt, have a long strike-slip history since the Mesozoic (Yin & Nie 1993; Dirks *et al.* 1994; Li 1994; Meng *et al.* 2005; Li *et al.* 2016).

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Therefore, a new model is proposed herein that can probably explain the preceding reviewed dispute on the strike-slip distance along the ASRRF.

7 CONCLUSIONS

Combining with U-Pb SIMS dating on zircons, rock magnetic and petrographic analyses, we provide a volcanic-derived palaeomagnetic data set from the central Thailand. It indicates a palaeolatitude, for the Indochina Block, of 26.4 ± 11.8°N for the study area (15.25°N, 101.35°E) at ∼205 Ma, which is similar to the data obtained from the Huai Hin Lat Formation limestones and sandstones. Thus, no inclination shallowing bias is recognized from the previous reported palaeomagnetic data from the coeval sedimentary rocks. The consistence of palaeomagnetic data from both volcanic and sedimentary rocks confirms that the estimate of the Late Triassic position of Indochina is reliable. The Indochina, as well as the Northern Qiangtang Blocks have collided with the southern margin of Eurasia by the Late Triassic.

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Supplementary data are available at *GJI* online.

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